EDC PRESENTATION

DIODE MODELS

Ideal • Practical • Complete

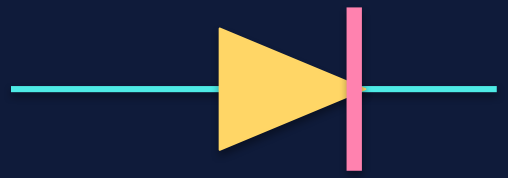
S. Ram Charan

Roll No. 26311A05JX

FORWARD BIAS & REVERSE BIAS

FORWARD BIAS

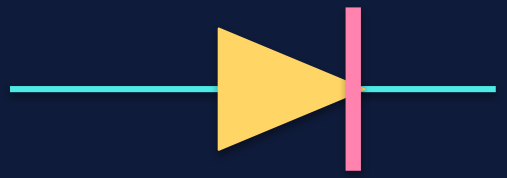
- P-side is connected to the positive terminal.
- N-side is connected to the negative terminal.
- The depletion region becomes narrower.
- Current flows easily after the cut-in voltage (about 0.7 V for silicon).



Forward direction →

REVERSE BIAS

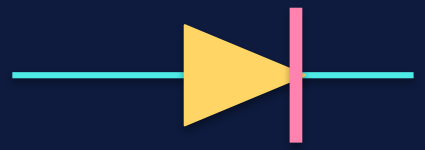
- P-side is connected to the negative terminal.
- N-side is connected to the positive terminal.
- The depletion region becomes wider.
- Only a tiny leakage current flows until breakdown.



Reverse bias: blocks current

THREE TYPES OF DIODE MODELS

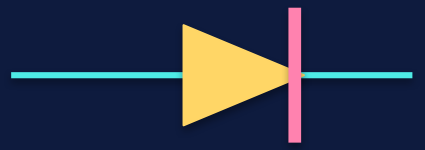
1. Ideal model



Perfect ON/OFF switch

ON = short circuit
OFF = open circuit

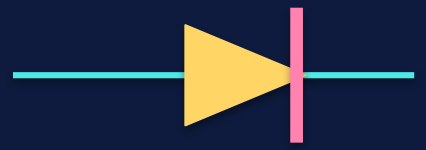
2. Simplified (Practical) model



Includes a fixed forward drop

$V_D \approx 0.7$ V (silicon)

3. Piecewise-linear model



Adds internal/dynamic resistance

$V_D \approx V_\gamma + I_D r_D$

IDEAL DIODE MODEL — DEFINITION & FIGURE

DEFINITION

The diode is treated as a perfect switch: it has zero voltage drop when ON and blocks all current when OFF.

MODEL IDEA

ON: closed switch | OFF: open switch

ON: closed switch | OFF: open switch

EQUIVALENT CIRCUIT



Remember: models simplify real diode behaviour to make circuit calculations easier.

IDEAL DIODE MODEL — FORMULAS

KEY RELATIONSHIPS

- Forward ON: $V_D = 0$ V
- Forward ON: $I_D > 0$ (limited by external circuit)
- Reverse OFF: $I_D = 0$ A
- Ideal resistance: $R_D(ON) = 0\ \Omega$; $R_D(OFF) \to \infty$

QUICK CHECK

Ideal ON:
$V_D = 0$ V

Ideal OFF:
$I_D = 0$ A

Perfect switch assumption

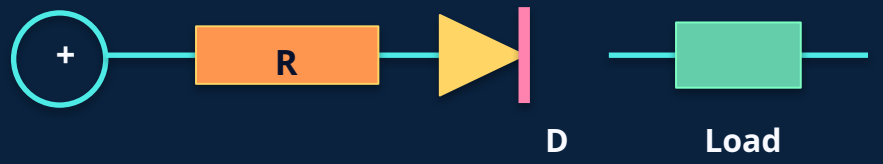
IDEAL DIODE MODEL — USES & APPLICATIONS

WHERE IT IS USEFUL

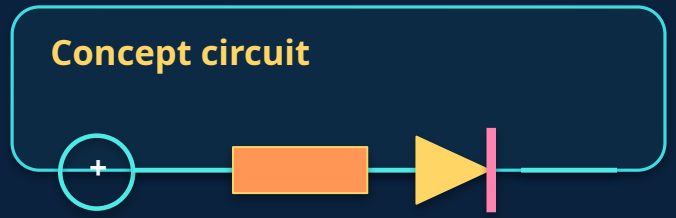
- Quick first-pass circuit analysis
- Understanding rectifier operation
- Switching logic and introductory circuit problems

Application models are approximations; choose the model based on the accuracy needed.

APPLICATION DIAGRAM



A simple series diode switch: the ideal model makes the ON path a short and the OFF path an open circuit.



SIMPLIFIED (PRACTICAL) DIODE MODEL — DEFINITION & ~~FIGURE~~

DEFINITION

The practical model includes a fixed threshold (cut-in) voltage to represent the diode's forward barrier.

MODEL IDEA

Silicon diode: fixed drop ≈ 0.7 V

$V_\gamma \approx 0.7$ V for silicon

EQUIVALENT CIRCUIT



$V_D \approx 0.7$ V

Remember: models simplify real diode behaviour to make circuit calculations easier.

SIMPLIFIED (PRACTICAL) DIODE MODEL — FORMULAS

KEY RELATIONSHIPS

- Forward ON: $V_D \approx V_\gamma$
- For silicon: $V_\gamma \approx 0.7$ V (approx.)
- Series circuit: $I_D \approx (V_s - V_\gamma)/R$
- Reverse OFF (before breakdown): $I_D \approx 0$ A

QUICK CHECK

Example
$V_s = 5$ V
R = 1 kΩ
$V_D \approx 0.7$ V

$I_D \approx (5 - 0.7)/1000$
$= 4.3$ mA

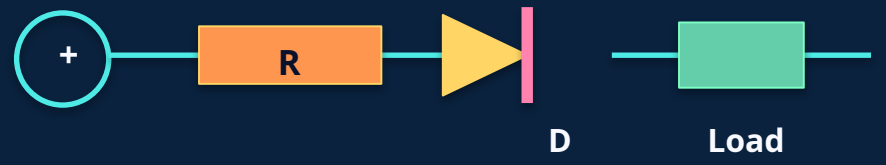
SIMPLIFIED (PRACTICAL) DIODE MODEL — USES & APPLICATIONS

WHERE IT IS USEFUL

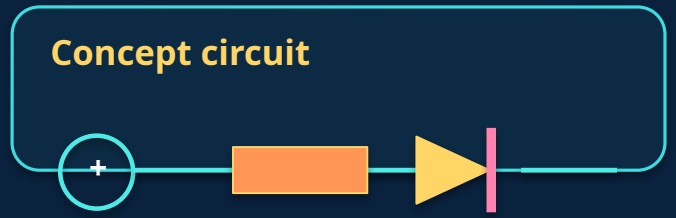
- Estimating LED/diode series current (use the relevant diode drop)
- Rectifier voltage estimates
- Power-supply and clamp-circuit calculations

Application models are approximations; choose the model based on the accuracy needed.

APPLICATION DIAGRAM



A DC source and resistor feed a forward-biased silicon diode; the resistor takes the remaining voltage.



PIECEWISE-LINEAR DIODE MODEL — DEFINITION & FIGURE

DEFINITION

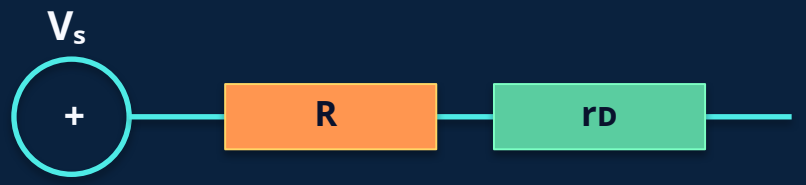
The piecewise-linear model includes a threshold voltage plus a small internal/dynamic resistance during conduction.

MODEL IDEA

Threshold source V_γ in series with r_D

V_γ and r_D are included

EQUIVALENT CIRCUIT



$V_\gamma + r_D$

Remember: models simplify real diode behaviour to make circuit calculations easier.

PIECEWISE-LINEAR DIODE MODEL — FORMULAS

KEY RELATIONSHIPS

- Forward ON: $V_D \approx V_\gamma + I_D r_D$
- Current: $I_D \approx (V_s - V_\gamma)/(R + r_D)$
- Small-signal dynamic resistance: $r_D \approx nV_t/I_D$
- At room temperature: $V_t \approx 25.9$ mV

QUICK CHECK

$$I_D \approx \frac{(V_s - V_\gamma)}{(R + r_D)}$$

Includes internal resistance

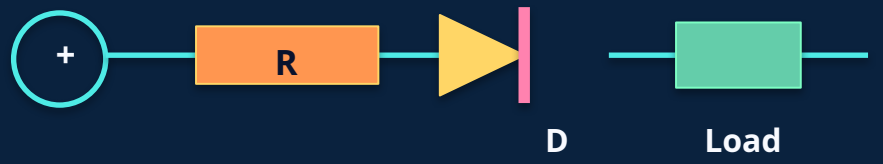
PIECEWISE-LINEAR DIODE MODEL — USES & APPLICATIONS

WHERE IT IS USEFUL

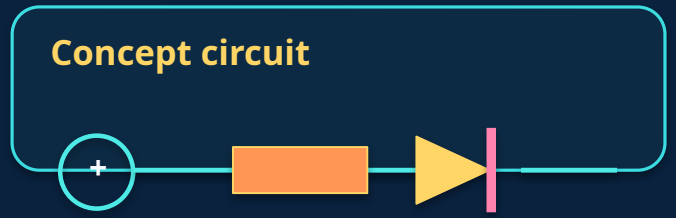
- More accurate hand analysis of forward-biased circuits
- Small-signal and load-line approximations
- Comparing current and voltage changes in diode circuits

Application models are approximations; choose the model based on the accuracy needed.

APPLICATION DIAGRAM



A source, external resistor and diode model with V_γ and r_D show why the diode voltage rises slightly as current increases.



SIGNING OFF

RC

THANK YOU!

Questions?

by~~
@...